Medical Staff Attrition Analysis

Group 11

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I. Executive Summary

This research utilizes healthcare personnel data transformed from the IBM Watson dataset, employing EDA and regression models to explore key factors affecting nursing staff turnover. The study reveals that compensation and promotion for entry-level employees have significant impacts on turnover rates, with job satisfaction and emotional factors also playing important roles. Finally, the research finds that workplace environment satisfaction remains a crucial indicator affecting employee turnover in healthcare institutions, providing specific recommendations for workplace environment enhancement to help medical institutions reduce personnel loss and improve staff stability.

II. Research Motivation

In recent years, the demand for healthcare personnel within the United States has continued to rise. According to research published by Mercer Consulting in 2021, the U.S. will face severe healthcare personnel shortages in the coming decade. The study indicates that by 2025, the U.S. will experience a shortage of over 400,000 home care workers and 29,400 specialized nurses.

Similar healthcare workforce shortages also appear in Taiwan. According to United Daily News reports, Taiwan's nursing workforce gap reached 16,000 to 24,000 people in 2023. These data show that whether in the U.S. or Taiwan, healthcare workforce shortages have become urgent challenges requiring solutions.

This research aims to identify key factors affecting nursing staff turnover through data analysis, deeply exploring how to reduce healthcare personnel turnover rates, and proposing improvement solutions for these factors, hoping to provide viable countermeasures for solving healthcare workforce shortage problems.

III. Data Sources

Kaggle dataset based on healthcare personnel data transformed from IBM Watson dataset: https://www.kaggle.com/datasets/jpmiller/employee-attrition-for-healthcare?select=watson_healthcare_modified.csv

Dataset Details:
Total of 35 columns, 1,676 data points. Fields include hourly wages, hospital departments,

distance from home, job satisfaction, and other indicators. Target variable Y is Attrition: 1 (left) accounts for 12%, 0 (stayed) accounts for 88%.

IV. Exploratory Data Analysis (EDA)

Direct observation of the data reveals several columns that can be removed. First are Over18 and Employee Count, as these two columns have no variation in values. Next is Employee ID, as these values are all unique.

First, we calculate correlation coefficients between different x variables to observe collinearity. Setting the correlation coefficient threshold at 0.6, we obtained the following results:

	Var1	Var2	Correlation
4	JobLevel	MonthlyIncome	0.9515721
7	PercentSalaryHike	PerformanceRating	0.7667374
8	Age	TotalWorkingYears	0.6925121
9	JobLevel	TotalWorkingYears	0.7809290
10	MonthlyIncome	TotalWorkingYears	0.7729383
12	TotalWorkingYears	YearsAtCompany	0.6224951
16	YearsAtCompany	YearsInCurrentRole	0.7594210
18	YearsAtCompany	YearsSinceLastPromotion	0.6169147
19	YearsAtCompany	YearsWithCurrManager	0.7712235
20	YearsInCurrentRole	YearsWithCurrManager	0.7215428

1. Model Data Preparation

- From dataset observation, many fields have only one value (such as EmployeeCount, Over18, StandardHours). Since these values provide little help for model interpretation, they are removed before building the model. (Figure 1)
- Remove EmployeeID as it's an employee number with no explanatory meaning. (Figure 1)
- Convert some variables to **categorical variables**, such as target variable Attrition, and independent variables Education, EnvironmentSatisfaction, JobInvolvement, etc. (Figure 2)
- Apply **log transformation** to independent variables X: After observation, many independent variables show **right-skewed distribution**, so log transformation is applied to the data. (Table 1)

Figure 1:	**Figure 2:**

```r
# 移除不需要的欄位
data <- data[, !(names(data) %in% c("EmployeeCount", "Over18", "StandardHours", "EmployeeID"))]
```

```r
# 確保 Attrition 為因子類型
data$Attrition <- as.factor(data$Attrition)

# education
data$Education <- as.factor(data$Education)

# environment_satisfaction
data$EnvironmentSatisfaction <- as.factor(data$EnvironmentSatisfaction)

# job involvement
data$JobInvolvement <- as.factor(data$JobInvolvement)

# job satisfaction
data$JobSatisfaction <- as.factor(data$JobSatisfaction)

# 將 JobLevel 轉換為因子變數
data$JobLevel <- as.factor(data$JobLevel)

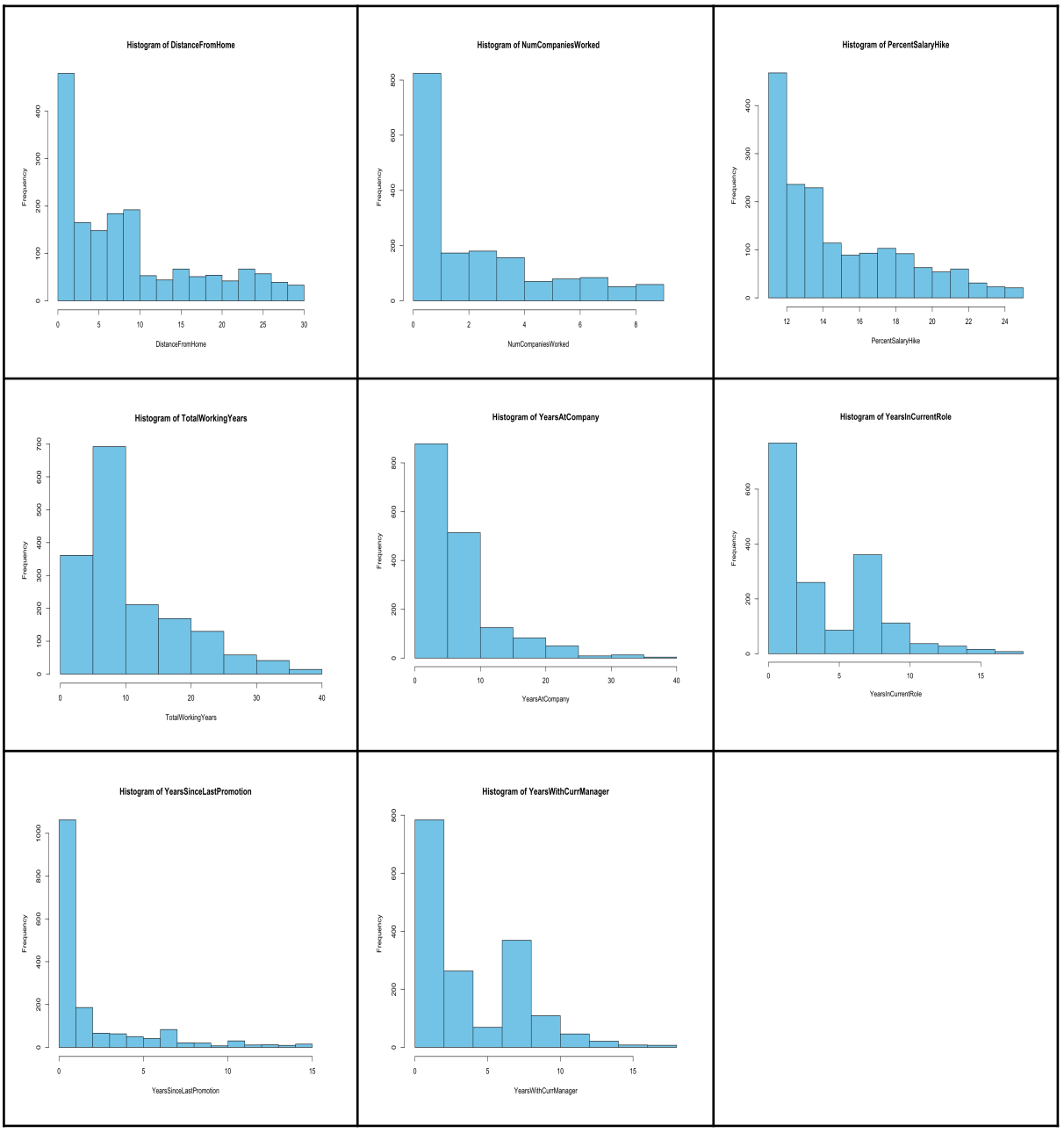
# 將 PerformanceRating 轉換為因子變數
data$PerformanceRating <- as.factor(data$PerformanceRating)

# relationship satisfaction
data$RelationshipSatisfaction <- as.factor(data$RelationshipSatisfaction)

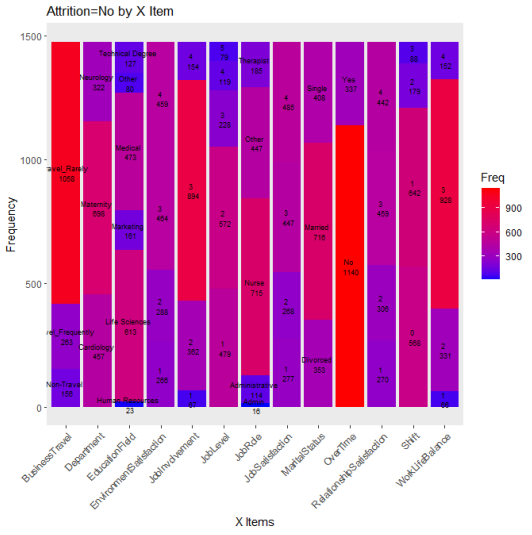
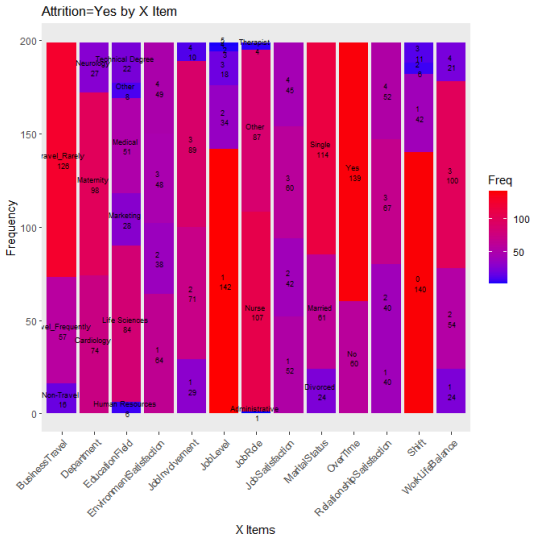
# shift
data$Shift <- as.factor(data$Shift)

# work life balance
data$WorkLifeBalance <- as.factor(data$WorkLifeBalance)
```

Table 1: Histogram of Each Variable



2. Distribution of Each Factor

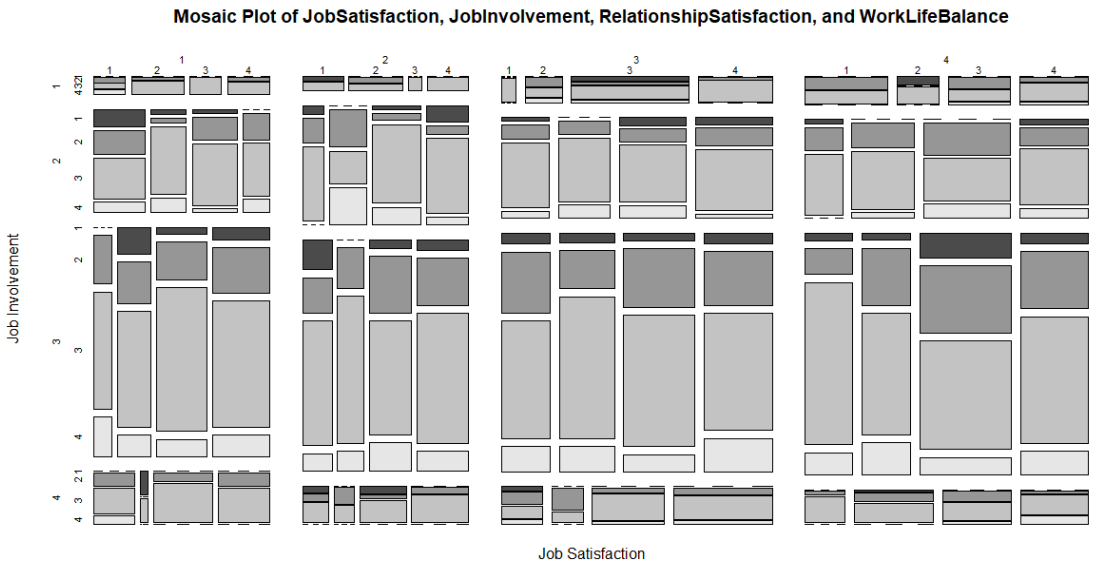
Chart #1 **When y-axis is attrition=no, x-axis shows distribution of different factor variables**	**Chart #2** **When y-axis is attrition=yes, x-axis shows distribution of different factor variables**
	

1. In BusinessTravel section, TravelFrequently shows higher proportion in Attrition=Yes
2. EnvironmentSatisfaction=1 shows slight increase in Attrition=Yes proportion
3. For Attrition=yes, JobInvolvement at levels 1 and 2 both show slight increases
4. JobLevel shows level 1 has higher proportion when Attrition=yes, while level 2 is higher for Attrition=No
5. For Attrition=yes, MaritalStatus shows more single individuals
6. Overtime shows more overtime work when Attrition=yes
7. For Attrition=yes, Shift=0 is most common

3. Mosaic Plot

Chart #3 Mosaic Plot of JobSatisfaction, JobInvolvement, RelationshipSatisfaction, and WorkLifeBalance

```
> ftable(xtabs(~JobSatisfaction+JobInvolvement+RelationshipSatisfaction+WorkLifeBalance, data=data), row.vars=1:2)
                                RelationshipSatisfaction  1           2           3           4
                                WorkLifeBalance           1  2  3  4  1  2  3  4  1  2  3  4  1  2  3  4
JobSatisfaction JobInvolvement
1               1                                         0  1  1  1  0  1  4  0  0  0  3  0  0  1  3  0
                2                                         5  7 12  3  1  1 14  3  1  6 16  1  0  4  8  2
                3                                         0  5 12  4  5  8 22  4  2 11 42  5  4 15 41  7
                4                                         0  3  6  2  1  0  1  0  0  3 13  0  0  4 10  0
2               1                                         1  0  2  0  0  2  2  0  0  0  1  0  0  0  3  0
                2                                         1  3  9  0  0  8  7  8  1  2 22  5  4  2 18  2
                3                                         5  6 21  3  0  6 22  3  2 14 28  7  3 14 38  6
                4                                         1  1  3  0  0  2  2  0  2  1  6  0  0  2  9  0
3               1                                         0  0  2  0  0  2  2  1  3  2  9  2  0  1  9  0
                2                                         1  4 18  2  0  4 19  4  3  5 22  5  3  8 27  2
                3                                         3 17 33  7  3 12 45  8  3 24 54  7  4 21 46 13
                4                                         1  3  3  1  0  4  2  0  0  2 11  1  0  4 13  2
4               1                                         0  6  6  0  2  0  4  0  0  4  4  1  0  2  7  1
                2                                         1  5 13  0  0  9 21  2  0 16 21  7  2  7 22  4
                3                                         2  7 44  6  2 16 34  9 13 35 57  9  4 22 49  9
                4                                         0  1  6  0  1  4  9  0  0  4  7  1  0  1  8  3
```



- From the Mosaic plot, most people are located between JobInvolvement=3 and JobSatisfaction=3/4
- Basically, JobSatisfaction, JobInvolvement, WorkLifeBalance, and RelationshipSatisfaction variables are independent of each other with no significant associations

Chart #4 Mosaic Plot of Shift, EnvironmentSatisfaction, JobSatisfaction, and RelationshipSatisfaction

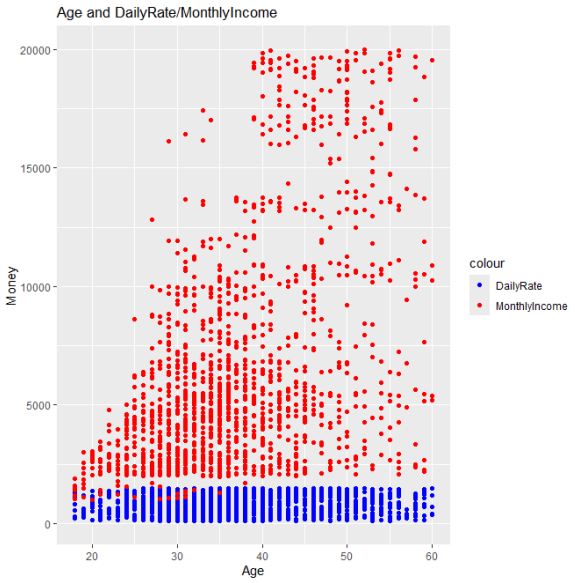
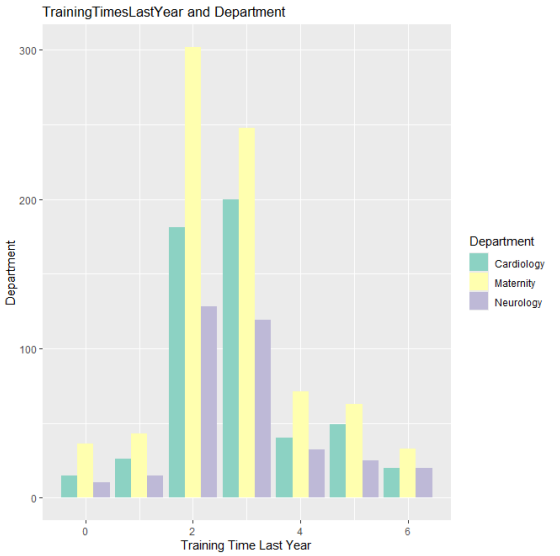
```
> ftable(xtabs(~Shift+EnvironmentSatisfaction+JobSatisfaction+RelationshipSatisfaction, data=data)
+ , row.vars=1:2)
                              JobSatisfaction           1           2           3           4
                              RelationshipSatisfaction  1  2  3  4  1  2  3  4  1  2  3  4  1  2  3  4
Shift EnvironmentSatisfaction
0     1                                                 8  5  9  6  6  6 11  7 10 11 16 17  6  6 20  7
      2                                                 4  5  4  8  5  6  6 10 12 12 16  8  7  7 15 12
      3                                                10  8 11 23  3  6 10 10  7  6 19 27  9 14 19 23
      4                                                 5 10 13 10  8  7 11 19 11 12 14 23 12 14 25 21
1     1                                                 9  1  6 11  0  2  7  7  5  9 15 13  7  6  8 14
      2                                                 3  9  9 10  3  7 10  7 11  7  7  9 12  8 13 10
      3                                                 5  7 16  8  8 11 10 16 16 12 20 16 12 26 31  9
      4                                                 7 10 18 12 11  5 12 10 12 18 18 16  8 15 19 15
2     1                                                 1  0  0  3  2  2  0  2  1  4  4  3  5  3  2  8
      2                                                 0  0  3  3  1  2  0  2  4  2  1  2  2  1  1  5
      3                                                 5  1  3  1  3  3  3  4  2  6  5  3  2  3  7  3
      4                                                 2  4  3  0  3  1  2  3  1  2  5 10  6  7  8  5
3     1                                                 0  0  0  1  1  0  0  2  1  1  3  3  3  1  2  1
      2                                                 0  3  4  1  1  1  2  2  0  2  3  0  0  1  3  2
      3                                                 2  2  2  2  1  1  6  0  2  2  1  2  5  0  0  2
      4                                                 1  0  2  0  0  2  1  0  0  0  6  1  1  1  6  4
```



- When Shift=0 and 1, all variables appear independent with little interaction
- When Shift=2 and 3, RelationshipSatisfaction=4 affects JobSatisfaction distribution
- When Shift=3 and EnvironmentSatisfaction=4, JobSatisfaction=2, 3, and 4 affect RelationshipSatisfaction distribution

4. Analysis of Age, DailyRate, MonthlyIncome, and TrainingTimesLastYear

Chart #5	**Chart #6**

Age, DailyRate, MonthlyIncome Relationships	**TrainingTimesLastYear by Different Departments**
	
1. From the table, employees under 25 don't have monthly salaries exceeding $5,000	1. Regardless of department, people with 2-3 training sessions are most common 2. Department quantities: Maternity > Cardiology > Neurology 3. People with 0 and 6 training sessions are least common

V. Model

1. Original Model

logistic_model <- glm(Attrition ~ ., data = data, family = "binomial")

Variable	Target variable y (Attrition) processed without modification.

Summary

```
Coefficients:
                                 Estimate Std. Error z value Pr(>|z|)
(Intercept)                     8.599e-01  6.538e+02   0.001 0.998951
Age                            -9.953e-02  2.280e-02  -4.366 1.27e-05 ***
BusinessTravelTravel_Frequently 2.039e+00  6.222e-01   3.277 0.001050 **
BusinessTravelTravel_Rarely     2.631e-01  5.557e-01   0.474 0.635844
DailyRate                      -8.606e-04  3.551e-04  -2.424 0.015365 *
DepartmentMaternity            -1.634e+00  4.221e-01  -3.870 0.000109 ***
DepartmentNeurology            -1.717e+00  5.490e-01  -3.127 0.001767 **
Education2                     -2.237e-01  5.277e-01  -0.424 0.671667
Education3                      2.090e-01  4.611e-01   0.453 0.650376
Education4                      6.624e-01  5.052e-01   1.311 0.189779
Education5                     -1.720e+00  1.280e+00  -1.344 0.179057
EducationFieldLife Sciences    -3.658e+00  1.223e+00  -2.992 0.002774 **
EducationFieldMarketing        -2.853e+00  1.297e+00  -2.199 0.027878 *
EducationFieldMedical          -3.733e+00  1.220e+00  -3.059 0.002223 **
EducationFieldOther            -3.555e+00  1.380e+00  -2.576 0.009993 **
EducationFieldTechnical Degree -2.274e+00  1.254e+00  -1.814 0.069749 .
EnvironmentSatisfaction2       -1.827e+00  4.502e-01  -4.057 4.96e-05 ***
EnvironmentSatisfaction3       -2.413e+00  4.437e-01  -5.439 5.37e-08 ***
EnvironmentSatisfaction4       -2.602e+00  4.249e-01  -6.123 9.17e-10 ***
GenderMale                      1.559e-02  2.935e-01   0.053 0.957630
HourlyRate                      3.442e-03  7.651e-03   0.450 0.652837
JobInvolvement2                -1.974e+00  5.540e-01  -3.563 0.000367 ***
JobInvolvement3                -3.130e+00  5.384e-01  -5.814 6.11e-09 ***
JobInvolvement4                -4.353e+00  7.950e-01  -5.476 4.34e-08 ***
JobLevel2                      -2.850e+00  6.590e-01  -4.324 1.53e-05 ***
JobLevel3                       1.096e-01  1.045e+00   0.105 0.916459
JobLevel4                      -9.132e-02  1.912e+00  -0.048 0.961901
JobLevel5                       2.521e+00  2.414e+00   1.044 0.296439
JobRoleAdministrative           1.100e+01  6.538e+02   0.017 0.986581
JobRoleNurse                    1.294e+01  6.538e+02   0.020 0.984207
JobRoleOther                    1.175e+01  6.538e+02   0.018 0.985657
JobRoleTherapist                1.180e+01  6.538e+02   0.018 0.985606
JobSatisfaction2               -7.765e-01  4.465e-01  -1.739 0.082014 .
JobSatisfaction3               -9.001e-01  3.828e-01  -2.351 0.018710 *
JobSatisfaction4               -2.014e+00  4.338e-01  -4.644 3.42e-06 ***
MaritalStatusMarried            5.079e-01  4.572e-01   1.111 0.266631
MaritalStatusSingle             1.210e+00  6.110e-01   1.980 0.047735 *
MonthlyIncome                  -2.184e-04  1.407e-04  -1.552 0.120740
MonthlyRate                     2.679e-05  2.036e-05   1.316 0.188237
OverTimeYes                     5.385e+00  4.629e-01  11.635  < 2e-16 ***
PerformanceRating4             -1.612e-01  6.171e-01  -0.261 0.793880
RelationshipSatisfaction2      -1.127e+00  4.922e-01  -2.290 0.022029 *
RelationshipSatisfaction3      -1.106e+00  4.261e-01  -2.594 0.009473 **
RelationshipSatisfaction4      -1.136e+00  4.251e-01  -2.672 0.007536 **
Shift1                         -2.372e+00  5.285e-01  -4.489 7.17e-06 ***
Shift2                         -1.967e+00  7.483e-01  -2.629 0.008561 **
Shift3                         -1.263e+00  7.209e-01  -1.752 0.079717 .
TrainingTimesLastYear          -2.744e-01  1.188e-01  -2.309 0.020927 *
WorkLifeBalance2               -1.834e+00  5.286e-01  -3.469 0.000522 ***
WorkLifeBalance3               -3.410e+00  5.396e-01  -6.319 2.62e-10 ***
WorkLifeBalance4               -2.627e+00  6.695e-01  -3.924 8.70e-05 ***
Log_DistanceFromHome            1.275e+00  1.992e-01   6.399 1.56e-10 ***
Log_NumCompaniesWorked          1.584e+00  2.788e-01   5.681 1.34e-08 ***
Log_PercentSalaryHike           3.258e-02  9.892e-01   0.033 0.973726
Log_TotalWorkingYears          -1.471e+00  4.025e-01  -3.655 0.000257 ***
Log_YearsAtCompany             -1.588e-01  5.105e-01  -0.311 0.755751
Log_YearsInCurrentRole         -5.565e-01  3.688e-01  -1.509 0.131317
Log_YearsSinceLastPromotion     8.718e-01  2.594e-01   3.361 0.000776 ***
Log_YearsWithCurrManager       -7.036e-01  3.206e-01  -2.195 0.028160 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1221.46  on 1675  degrees of freedom
Residual deviance:  386.37  on 1617  degrees of freedom
AIC: 504.37

Number of Fisher Scoring iterations: 16
```

Chi-square Test

```r
# 建立空模型（只有截距項）
null_model <- glm(Attrition ~ 1, data = data, family = "binomial")

# 提取空模型的 deviance 和自由度
null_deviance <- null_model$deviance
null_df <- null_model$df.residual

# 計算卡方統計量和自由度
chi_squared_original <- null_deviance - logistic_model$deviance
df_original <- null_df - logistic_model$df.residual

# 計算 p-value
p_value_original <- pchisq(chi_squared_original, df = df_original, lower.tail = FALSE)

# 輸出結果
cat("原始模型 vs 空模型:\n")
cat("卡方統計量 (Chi-squared):", chi_squared_original, "\n")
cat("自由度差異 (df):", df_original, "\n")
cat("p-value:", p_value_original, "\n")

# 判斷是否顯著
if (p_value_original < 0.05) {
  cat("結果：顯著\n")
} else {
  cat("結果：不顯著\n")
}
```

	原始模型 vs 空模型:
	> cat("卡方統計量 (Chi-squared):", chi_squared_original, "\n")
	卡方統計量 (Chi-squared): 835.0843
	> cat("自由度差異 (df):", df_original, "\n")
	自由度差異 (df): 58
	> cat("p-value:", p_value_original, "\n")
	p-value: 3.884077e-138
	> # 判斷是否顯著
	> if (p_value_original < 0.05) {
	+ cat("結果：顯著\n")
	+ } else {
	+ cat("結果：不顯著\n")
	+ }
	結果：顯著

2. Bidirectional Stepwise Regression on Original Model

best_model <- step(logistic_model, direction = "both", trace = 1)

Variable	Target variable y (Attrition) processed without modification.

Summary

```
Coefficients:
                                Estimate Std. Error z value Pr(>|z|)
(Intercept)                    9.940e-01  6.812e+02   0.001 0.998836
Age                           -9.363e-02  2.190e-02  -4.276 1.90e-05 ***
BusinessTravelTravel_Frequently  2.231e+00  6.012e-01   3.711 0.000207 ***
BusinessTravelTravel_Rarely     4.455e-01  5.340e-01   0.834 0.404115
DailyRate                     -8.569e-04  3.400e-04  -2.520 0.011725 *
DepartmentMaternity           -1.691e+00  4.105e-01  -4.119 3.81e-05 ***
DepartmentNeurology           -1.689e+00  5.446e-01  -3.102 0.001922 **
EducationFieldLife Sciences   -3.419e+00  1.166e+00  -2.932 0.003367 **
EducationFieldMarketing       -2.606e+00  1.240e+00  -2.101 0.035598 *
EducationFieldMedical         -3.554e+00  1.160e+00  -3.064 0.002185 **
EducationFieldOther           -3.286e+00  1.306e+00  -2.516 0.011867 *
EducationFieldTechnical Degree  -2.225e+00  1.204e+00  -1.847 0.064687 .
EnvironmentSatisfaction2      -1.902e+00  4.401e-01  -4.322 1.54e-05 ***
EnvironmentSatisfaction3      -2.427e+00  4.309e-01  -5.633 1.77e-08 ***
EnvironmentSatisfaction4      -2.658e+00  4.162e-01  -6.387 1.69e-10 ***
JobInvolvement2               -1.857e+00  5.271e-01  -3.523 0.000426 ***
JobInvolvement3               -2.950e+00  5.102e-01  -5.782 7.40e-09 ***
JobInvolvement4               -4.170e+00  7.633e-01  -5.463 4.68e-08 ***
JobLevel2                     -2.852e+00  6.354e-01  -4.489 7.17e-06 ***
JobLevel3                     -1.207e-03  1.002e+00  -0.001 0.999039
JobLevel4                     -1.988e-01  1.809e+00  -0.110 0.912507
JobLevel5                      2.647e+00  2.322e+00   1.140 0.254216
JobRoleAdministrative          9.914e+00  6.812e+02   0.015 0.988388
JobRoleNurse                   1.265e+01  6.812e+02   0.019 0.985184
JobRoleOther                   1.153e+01  6.812e+02   0.017 0.986492
JobRoleTherapist               1.166e+01  6.812e+02   0.017 0.986337
JobSatisfaction2              -7.494e-01  4.331e-01  -1.730 0.083582 .
JobSatisfaction3              -8.275e-01  3.715e-01  -2.227 0.025926 *
JobSatisfaction4              -1.964e+00  4.215e-01  -4.660 3.17e-06 ***
MaritalStatusMarried           4.896e-01  4.319e-01   1.134 0.256967
MaritalStatusSingle            1.316e+00  5.909e-01   2.227 0.025940 *
MonthlyIncome                 -1.959e-04  1.360e-04  -1.441 0.149632
OverTimeYes                    5.342e+00  4.512e-01  11.839  < 2e-16 ***
RelationshipSatisfaction2     -1.169e+00  4.698e-01  -2.489 0.012811 *
RelationshipSatisfaction3     -9.958e-01  4.063e-01  -2.451 0.014238 *
RelationshipSatisfaction4     -1.055e+00  4.119e-01  -2.562 0.010421 *
Shift1                        -2.175e+00  5.050e-01  -4.307 1.66e-05 ***
Shift2                        -1.974e+00  7.229e-01  -2.731 0.006315 **
Shift3                        -1.225e+00  6.948e-01  -1.763 0.077983 .
TrainingTimesLastYear         -2.644e-01  1.141e-01  -2.317 0.020498 *
WorkLifeBalance2              -1.847e+00  5.125e-01  -3.603 0.000314 ***
WorkLifeBalance3              -3.380e+00  5.268e-01  -6.415 1.41e-10 ***
WorkLifeBalance4              -2.545e+00  6.535e-01  -3.895 9.81e-05 ***
Log_DistanceFromHome           1.270e+00  1.916e-01   6.626 3.45e-11 ***
Log_NumCompaniesWorked         1.604e+00  2.635e-01   6.088 1.14e-09 ***
Log_TotalWorkingYears         -1.561e+00  3.759e-01  -4.154 3.27e-05 ***
Log_YearsInCurrentRole        -4.717e-01  2.791e-01  -1.690 0.090969 .
Log_YearsSinceLastPromotion    8.435e-01  2.476e-01   3.406 0.000659 ***
Log_YearsWithCurrManager      -7.939e-01  2.722e-01  -2.916 0.003541 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1221.46  on 1675  degrees of freedom
Residual deviance:  395.99  on 1627  degrees of freedom
AIC: 493.99

Number of Fisher Scoring iterations: 16
```

Chi-square Test

```r
# 計算卡方統計量和自由度
chi_squared_stepwise <- null_deviance - best_model$deviance
df_stepwise <- null_df - best_model$df.residual

# 計算 p-value
p_value_stepwise <- pchisq(chi_squared_stepwise, df = df_stepwise, lower.tail = FALSE)

# 輸出結果
cat("\n逐步回歸模型 vs 空模型:\n")
cat("卡方統計量 (Chi-squared):", chi_squared_stepwise, "\n")
cat("自由度差異 (df):", df_stepwise, "\n")
cat("p-value:", p_value_stepwise, "\n")

# 判斷是否顯著
if (p_value_stepwise < 0.05) {
  cat("結果：顯著\n")
} else {
  cat("結果：不顯著\n")
}
```

```
逐步回歸模型 vs 空模型:
> cat("卡方統計量 (Chi-squared):", chi_squared_stepwise, "\n")
卡方統計量 (Chi-squared): 825.4666
> cat("自由度差異 (df):", df_stepwise, "\n")
自由度差異 (df): 48
> cat("p-value:", p_value_stepwise, "\n")
p-value: 3.349195e-142
> # 判斷是否顯著
> if (p_value_stepwise < 0.05) {
+   cat("結果：顯著\n")
+ } else {
+   cat("結果：不顯著\n")
+ }
結果：顯著
```

We performed bidirectional Stepwise Regression (including Backward Elimination and Forward Selection) to select variables, finding that variables with significant effects for predicting Attrition can be selected as the following:

Age, BusinessTravel, DailyRate, Department, DistanceFromHome, EducationField, EnvironmentSatisfaction, JobInvolvement, JobLevel, JobRole, JobSatisfaction, MaritalStatus, MonthlyIncome, OverTime, RelationshipSatisfaction, Shift, TrainingTimesLastYear, WorkLifeBalance, log(YearsInCurrentRole), log(YearsSinceLastPromotion), log(YearsWithCurrManager), log(DistanceFromHome), log(NumCompaniesWorked), log(TotalWorkingYears)

Theme 1: Do Salary Increases and Promotions Have Different Effects on Reducing Turnover Rates Across Different Job Levels? Does the Company Have an Optimal Salary Structure?

Adjusting healthcare personnel salaries and providing promotions are the most direct changes companies can make. In the basic model, we observe that JobLevel shows significant relationship with turnover rates. Although MonthlyIncome is not significant, we infer from its high collinearity with JobLevel that MonthlyIncome might have different effects on turnover across different JobLevels. Therefore, we want to verify whether **salary levels are related to turnover rates across all job levels, and whether promotions at each level are also related to turnover rates.**

First, to verify whether MonthlyIncome affects turnover rates differently across JobLevels, we decided to divide the data into five groups by JobLevel and perform logistic regression using the basic model excluding JobLevel variable. Following are partial regression results for Level1:

```
MonthlyIncome                -1.137e-03  3.190e-04  -3.563 0.000367 ***
MonthlyRate                   5.256e-06  3.049e-05   0.172 0.863140
OverTimeYes                   6.425e+00  7.580e-01   8.476  < 2e-16 ***
RelationshipSatisfaction2    -2.534e+00  7.457e-01  -3.398 0.000679 ***
RelationshipSatisfaction3    -2.357e+00  6.370e-01  -3.701 0.000215 ***
RelationshipSatisfaction4    -2.935e+00  7.101e-01  -4.133 3.57e-05 ***
Shift1                       -2.001e+00  7.979e-01  -2.508 0.012151 *
Shift2                       -2.625e+00  1.226e+00  -2.141 0.032248 *
Shift3                       -8.838e-01  1.003e+00  -0.881 0.378103
TrainingTimesLastYear        -1.412e-01  1.649e-01  -0.856 0.391880
WorkLifeBalance2             -3.341e+00  8.034e-01  -4.159 3.19e-05 ***
WorkLifeBalance3             -4.206e+00  8.173e-01  -5.146 2.66e-07 ***
WorkLifeBalance4             -2.687e+00  8.950e-01  -3.002 0.002678 **
Log_DistanceFromHome          1.393e+00  2.993e-01   4.656 3.22e-06 ***
Log_NumCompaniesWorked        1.797e+00  4.165e-01   4.316 1.59e-05 ***
Log_TotalWorkingYears        -1.568e+00  5.819e-01  -2.695 0.007039 **
Log_YearsInCurrentRole        1.160e-01  4.845e-01   0.239 0.810821
Log_YearsSinceLastPromotion   8.008e-01  4.384e-01   1.827 0.067726 .
Log_YearsWithCurrManager     -1.192e+00  4.253e-01  -2.802 0.005074 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)
```

We found that at Level 1, applying the Basic model clearly shows salary negatively correlated with turnover rates. For Level 2+ data, due to insufficient turnover numbers, the algorithm cannot converge significantly. Therefore, we reduced model variables and found through multiple tests that Level 2, Level 3, Level 4, and Level 5 all show no significant effects.

Therefore, we can conclude that **salary amount only affects turnover rates of entry-level employees. For Level 2+ employees, salary amount is not a reason for turnover. Therefore, company compensation policies should focus more on Level 1.**

Next, to verify whether promotion has significant effects on turnover rates across different levels, we divided the data into four groups covering Level 1&2, Level 2&3, Level 3&4, and Level 4&5, performing logistic regression on Attrition using the basic model (this time including JobLevel).

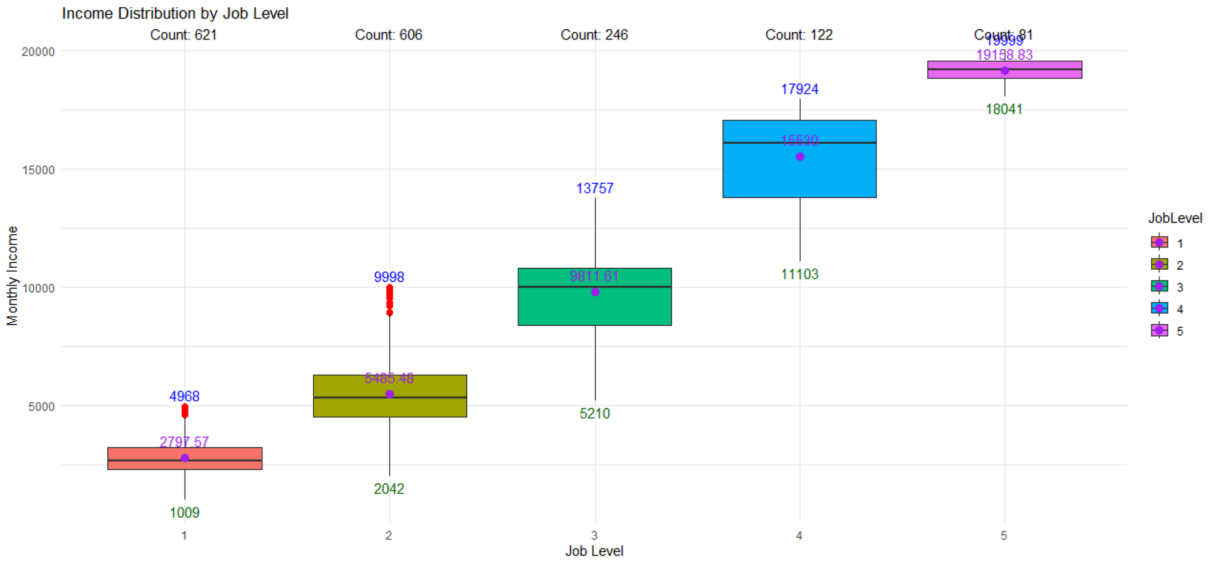
Partial results for Level 1 and Level 2 data:

JobLevel2	-2.931e+00	7.360e-01	-3.983	6.82e-05	***
JobRoleOther	-1.358e+00	3.923e-01	-3.463	0.000535	***
JobRoleTherapist	-2.656e-01	1.111e+00	-0.239	0.811061	
JobSatisfaction2	-9.285e-01	5.123e-01	-1.812	0.069942	.
JobSatisfaction3	-9.325e-01	4.379e-01	-2.129	0.033227	*
JobSatisfaction4	-2.348e+00	4.940e-01	-4.754	2.00e-06	***
MaritalStatusMarried	3.365e-01	4.923e-01	0.684	0.494195	
MaritalStatusSingle	1.555e+00	6.957e-01	2.236	0.025374	*
MonthlyIncome	-3.941e-04	1.806e-04	-2.182	0.029074	*
MonthlyRate	9.287e-06	2.307e-05	0.403	0.687273	
OverTimeYes	5.840e+00	5.520e-01	10.579	< 2e-16	***
RelationshipSatisfaction2	-9.713e-01	5.264e-01	-1.845	0.065008	.
RelationshipSatisfaction3	-9.371e-01	4.683e-01	-2.001	0.045397	*
RelationshipSatisfaction4	-1.214e+00	4.799e-01	-2.529	0.011452	*
Shift1	-2.407e+00	6.201e-01	-3.882	0.000104	***
Shift2	-1.835e+00	8.615e-01	-2.130	0.033150	*
Shift3	-1.603e+00	8.215e-01	-1.951	0.051027	.
TrainingTimesLastYear	-2.237e-01	1.295e-01	-1.728	0.083961	.
WorkLifeBalance2	-1.934e+00	5.879e-01	-3.289	0.001005	**
WorkLifeBalance3	-3.491e+00	6.160e-01	-5.668	1.44e-08	***

Results show that Level 2 compared to Level 1 has significantly reduced turnover rates. Using other datasets for regression also shows that Level 3 compared to Level 2 has significantly increased turnover rates, while Level 3&4 and Level 4&5 data also show algorithm convergence issues due to low turnover numbers, so only partial variables with JobLevel were included in regression, finding no significance in these two datasets.

Therefore, we can conclude that **promoting employees from Level 1 to Level 2 reduces turnover rates, but promotion from Level 2 to Level 3 may increase turnover rates, possibly due to more job-hopping opportunities at that level.** Therefore, when promoting employees from Level 2 to Level 3, careful evaluation is needed to ensure the employee has sufficient company identification, belonging, and satisfaction.

We further observe the Level 1&2 model to see whether promotion or salary increase is better for retaining entry-level employees. Assuming fixed marginal effects, we observe coefficients: JobLevel coefficient is -2.93, while salary coefficient is -0.000394. To achieve the same retention effect as JobLevel, salary increase would need to be -2.93/-0.000394, approximately $7,436 monthly income. For entry-level employees, this is impossible. **Therefore, if Level 2 positions are available, promotion is definitely better than salary increases for reducing turnover rates.**



Finally, regarding compensation management policy, we want to confirm whether current hospital entry-level employee salaries have reached optimal levels. We can apply financial management optimal capital structure concepts to see if the marginal cost of salary increases exceeds the expected marginal cost reduction from decreased turnover odds (replacement and search costs).

According to literature, employee turnover total cost is approximately 200% of annual salary. Our data shows entry-level employees average $2,797.57 monthly, turnover rate is 0.229, so total turnover cost is 2797.57×2×12 = 67,414.68. Assuming 5% annual interest rate, increasing monthly salary by $1 as $12 annual perpetuity increases company NPV payment by 12/0.05 = $240.

The odds ratio increase comes from Level 1 regression MonthlyIncome coefficient: exp(-0.001137) = 0.9988636 times original odds ratio. Solving equation (0.229+x)/(0.771-x) = 0.9988636×(0.229/0.771), we get x = -0.0002, meaning turnover probability decreases by 0.0002, saving 0.002×67,414.68 = $13.483. This shows salary increase costs are still higher, so there's no need for salary increases; salary reduction direction would have optimal salary.

What is the optimal salary? Assuming all marginal costs don't increase, we want turnover rate increase ratio r×67,414.68 to equal money saved from reducing α salary units: 240α = r×67,414.68

The odds ratio becomes exp(0.001137×α) times, so: (0.229+r)/(0.771-r) = exp(0.001137×α)×(0.229/0.771)

Solving this binary equation system yields no solution. This means **if assuming fixed marginal costs, the best approach is to give healthcare employees the lowest possible salary.** Of course, real-world marginal effects will increase, so reality must have an optimal solution. From this example, we can understand why governments need to set minimum wages, as many companies' optimal salaries might be extremely low.

Theme 2: Job Satisfaction is Significantly Related to Turnover. What Factors Have the Greatest Impact on Job Satisfaction?

Base Model	**Model Description:** Based on Best Model findings that job satisfaction is highly correlated with turnover rates, we further explore what factors have the greatest impact on job satisfaction, selecting employee emotional satisfaction factors related to this issue as the Base Model for subsequent interaction term testing.

```
new_base_model <- glm(

Attrition ~ JobSatisfaction + OverTime + WorkLifeBalance + Age +
EnvironmentSatisfaction + RelationshipSatisfaction + Log_DistanceFromHome +
Log_YearsWithCurrManager + Log_NumCompaniesWorked + Log_TotalWorkingYears +
TrainingTimesLastYear,

data = data,

family = "binomial")
```

Interaction Model 1	**Model Description:** After determining core variables, we further select significant variables to explore possible interactions based on significant variables: 1. WorkLifeBalance and JobSatisfaction: Employee satisfaction with work-life balance may affect their job satisfaction perception 2. EnvironmentSatisfaction and RelationshipSatisfaction: Work environment satisfaction may affect employee interpersonal relationship experience 3. Log_NumCompaniesWorked and Log_TotalWorkingYears: Both are work experience related variables; longer total work years may jointly affect turnover rates with job-hopping frequency 4. OverTime and Age: Age may affect employee acceptance of overtime, thereby affecting turnover rates

```
interaction_model1 <- glm(

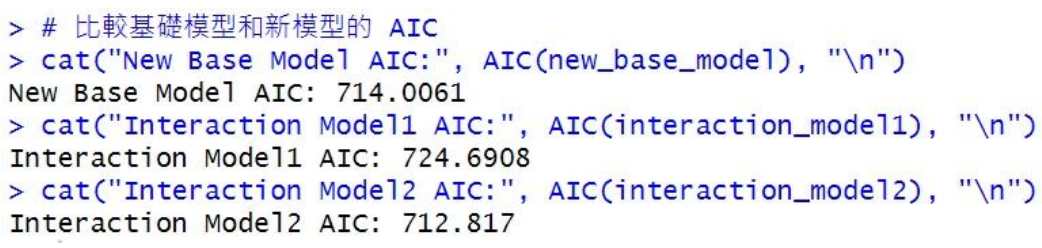
Attrition ~ JobSatisfaction * WorkLifeBalance + EnvironmentSatisfaction *
RelationshipSatisfaction + Log_NumCompaniesWorked * Log_TotalWorkingYears +
OverTime * Age + OverTime + WorkLifeBalance + JobSatisfaction + Age +
EnvironmentSatisfaction + RelationshipSatisfaction + Log_DistanceFromHome +
Log_YearsWithCurrManager + TrainingTimesLastYear,

data = data,

family = "binomial")
```

Next Step	Among interaction terms, only JobSatisfaction4:WorkLifeBalance2 and Log_NumCompaniesWorked:Log_TotalWorkingYears are significant, suggesting: ● High job satisfaction can mitigate turnover rates when work-life balance is low ● The combination of job-hopping frequency and work years affects turnover rates in certain situations Next, remove non-significant interaction terms to improve model interpretability

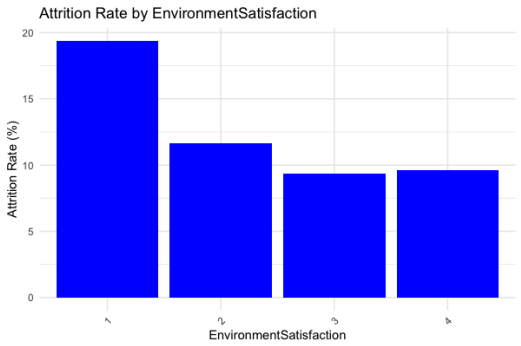
Interaction Model 2	interaction_model2 <- glm(Attrition ~ JobSatisfaction * WorkLifeBalance + EnvironmentSatisfaction + RelationshipSatisfaction + Log_NumCompaniesWorked * Log_TotalWorkingYears + OverTime + Age + WorkLifeBalance + JobSatisfaction + Log_DistanceFromHome + Log_YearsWithCurrManager + TrainingTimesLastYear, data = data, family = "binomial")

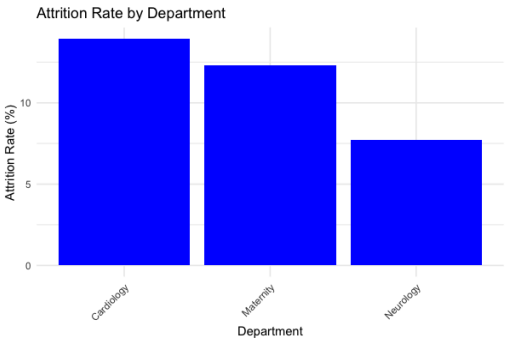
Conclusion	 Based on AIC, we select interaction_model2 as the final model, with improved interpretability and model simplicity, yielding the following management implications: 1. **Design exclusive growth opportunities for high job-hopping but senior employees to reduce turnover risk:** Looking at base_model alone, employee job-hopping frequency has no significant effect on turnover rates; and longer total work years significantly reduce turnover rates. But observing the interaction between the two reveals that employees with high job-hopping frequency and long tenure have higher turnover rates, so incentive mechanisms should be designed for senior employees to reduce turnover tendencies. 2. **Employee emotional satisfaction helps reduce turnover rates:** Emotional satisfaction covers multiple dimensions including work environment, interpersonal relationships, distance from home, and overtime. Employers should strive to create good work environments and provide timely care and assistance to enhance workplace satisfaction and belonging, further strengthening employee retention and job stability.

Theme 3: Verifying the Impact of Environmental Satisfaction on Turnover Rates Across Different Departments

Work environment quality directly affects employees. Assuming different departments have different work environments, we want to verify whether satisfaction has different impacts on turnover rates across departments.

From the basic model, we find that as environmental satisfaction increases, turnover rates decrease. Meanwhile, employees in Maternity and Neurology departments have lower turnover rates than Cardiology. Therefore, we want to explore whether different departments place the same emphasis on environmental satisfaction.





From these two charts, we can see that besides model significance, environmental satisfaction score of 1 clearly has higher turnover rates than scores 2-4, and Cardiology has higher turnover rates than other departments.

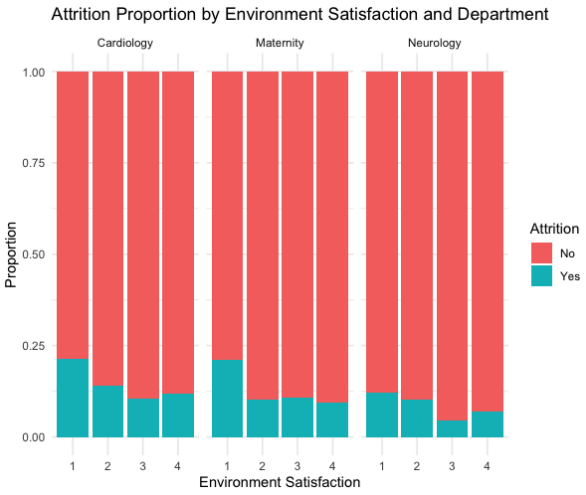
Next, we create contingency tables and turnover rate charts for departments and job satisfaction:

```
, ,  = No

                1   2   3   4
Cardiology     85 103 136 133
Maternity     123 132 222 221
Neurology      58  53 106 105

, ,  = Yes

                1   2   3   4
Cardiology     23  17  16  18
Maternity      33  15  27  23
Neurology       8   6   5   8
```



After chi-square testing, we found only the Maternity department is significant, indicating that for this department, environmental satisfaction affects turnover rates.

```
     Department    Chi_Square P_Value
[1,] "Cardiology"  6.868263   0.07621734
[2,] "Maternity"   14.28145   0.00254604
[3,] "Neurology"   3.959697   0.2658491
```

Next, we add interaction between EnvironmentSatisfaction and Department to the best_model, finding this interaction is not significant.

```
DepartmentMaternity:EnvironmentSatisfaction2 -4.351e-01  9.115e-01  -0.477 0.633170
DepartmentNeurology:EnvironmentSatisfaction2  1.529e-01  1.263e+00   0.121 0.903663
DepartmentMaternity:EnvironmentSatisfaction3  6.938e-01  8.122e-01   0.854 0.393013
DepartmentNeurology:EnvironmentSatisfaction3 -9.131e-01  1.289e+00  -0.708 0.478700
DepartmentMaternity:EnvironmentSatisfaction4  1.122e+00  8.186e-01   1.371 0.170434
DepartmentNeurology:EnvironmentSatisfaction4  5.465e-02  1.181e+00   0.046 0.963098
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1221.46  on 1675  degrees of freedom
Residual deviance:  390.43  on 1621  degrees of freedom
AIC: 500.43
```

ANOVA test:

```
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      1627     395.99
2      1621     390.43  6   5.5647   0.4737
```

Based on these results, we speculate that in the overall model, Maternity department's significance is diluted by other departments' data, plus possible insufficient sample size preventing detection of subtle interaction differences.

EnvironmentSatisfaction's impact within the Maternity department, though significant, has smaller global impact.

According to odds ratio calculations, employees with satisfaction level 2 have 82.6% reduced turnover odds, satisfaction level 3 employees have 93.3% reduced turnover odds, and satisfaction level 4 employees have 96.1% reduced turnover odds. Even from the above analysis showing no significant differences in turnover rates' response to work environment satisfaction across departments, work environment satisfaction remains an important indicator affecting employee turnover in healthcare institutions. Managers should strengthen provision of good work environments:

1. **Create friendly workplace culture:**
    - Encourage employee cooperation and communication, reduce stress from internal competition
    - Regularly hold employee care activities like birthday celebrations, family days, or appreciation events to enhance employee belonging
    - Implement transparent and fair performance evaluation systems and establish regular employee satisfaction assessment mechanisms
2. **Improve workplace facilities:**
    - Provide comfortable break spaces like lounges and tea rooms for employee relaxation during busy work

- Consider establishing gyms or sports facilities to encourage employees to maintain healthy physical condition and reduce stress
- Improve medical personnel work area design, such as providing ergonomic chairs and adjustable desks to ensure employee comfort during long working hours

VI. Conclusion

This research confirms the importance of compensation policies, promotion mechanisms, and emotional satisfaction for reducing healthcare personnel turnover rates. Salary impact is mainly concentrated on entry-level employees, while promotion is more effective than salary increases in reducing turnover rates. Job and environmental satisfaction are key to employee stability, especially for employees with high job-hopping frequency and long tenure. Employers should adopt multi-faceted strategies, including improving work environments, enhancing employee belonging, and implementing department-specific management measures to reduce turnover risk.

VII. Research Limitations

The dataset used in this research mainly comes from a specific platform (IBM Watson dataset), which may not comprehensively represent all healthcare institutions' situations.

VIII. References

1. https://www.techbang.com/posts/52880